

Table 1: Absolute trajectory error (ATE, cm) on RPG for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [3]	ESIO [3]	DEIO [4]	ES-PTAM [1]
rpg_box	1.92 ^a	4.31 ^b	4.41 ^b	11.38 ^b	N/R	4.06 ^b
rpg_monitor	0.91 ^a	2.31 ^b	3.48 ^b	7.87 ^b	N/R	2.34 ^b
rpg_bin	1.16 ^a	2.27 ^b	2.28 ^b	7.08 ^b	N/R	2.57 ^b
rpg_desk	1.46 ^a	1.57 ^b	2.03 ^b	3.16 ^b	N/R	2.84 ^b
rpg_reader	3.78 ^a	2.68 ^b	N/R	N/R	N/R	N/R

Sources: ^a[6], ^b[5].

Table 2: Absolute trajectory error (ATE, cm) on MVSEC for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [3]	ESIO [3]	DEIO [4]	ES-PTAM [1]
indoor_flying1	3.76 ^a	7.63 ^b	9.63 ^b	820.4 ^b	7.52 ^a	15.02 ^b
indoor_flying2	8.00 ^a	10.05 ^b	N/R	417.9 ^b	6.96 ^a	N/R
indoor_flying3	5.57 ^a	7.35 ^b	8.06 ^b	N/R	29.75 ^a	N/R
indoor_flying4	4.57 ^a	5.59 ^b	N/R	173.5 ^b	12.71 ^a	N/R

Sources: ^a[6], ^b[5].

References

- [1] FIXME — add the ES-PTAM reference (Ghosh et al.) by hand. Placeholder: this paper was never extracted into the results database, so no verified bibliographic data exists for it in this repository.
- [2] Yannick Burkhardt, Simon Schaefer, and Stefan Leutenegger. SuperEvent: Cross-modal learning of event-based keypoint detection. *arXiv preprint arXiv:2504.00139*, 2025.
- [3] Peiyu Chen, Weipeng Guan, and Peng Lu. ESVIO: Event-based stereo visual inertial odometry. *IEEE Robotics and Automation Letters*, 8(6):3661–3668, 2023.
- [4] Weipeng Guan, Fuling Lin, Peiyu Chen, and Peng Lu. DEIO: Deep event inertial odometry. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2411.03928.
- [5] Junkai Niu, Sheng Zhong, Xiuyuan Lu, Shaojie Shen, Guillermo Gallego, and Yi Zhou. ESVO2: Direct visual-inertial odometry with stereo event cameras. *IEEE Transactions on Robotics*, 2025. arXiv:2410.09374.
- [6] Junkai Zhong, Junkai Niu, and Yi Zhou. Stereo deep event visual odometry. *IEEE Robotics and Automation Letters*, 2025. arXiv:2509.08235.

Table 3: Absolute trajectory error (ATE, cm) on DSEC for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [3]	ESIO [3]	DEIO [4]	ES-PTAM [1]
city04_a	96.86 ^a	56.17 ^b	201.5 ^b	543.5 ^c	80.60 ^c	131.6 ^c
city04_b	17.65 ^a	73.83 ^b	48.33 ^b	295.1 ^c	35.40 ^c	29.02 ^c
city04_c	481.6 ^a	508.7 ^b	1400.8 ^b	896.2 ^c	413.8 ^c	1184.4 ^c
city04_d	406.2 ^a	546.6 ^b	921.7 ^c	2977.0 ^c	207.6 ^c	1053.9 ^c
city09_a	197.0 ^a	1183.0 ^a	328.5 ^a	N/R	N/R	N/R
city09_b	267.8 ^a	87.83 ^b	3481.5 ^a	2887.8 ^a	N/R	195.1 ^b
city09_c	931.1 ^a	1648.3 ^a	8043.5 ^a	N/R	N/R	3673.8 ^a
city09_d	564.3 ^a	1920.4 ^a	1677.1 ^a	1766.2 ^a	N/R	N/R
city09_e	246.3 ^a	1075.1 ^a	438.2 ^a	4201.0 ^a	N/R	1480.8 ^a
city11_a	47.56 ^a	48.77 ^b	406.1 ^b	107.4 ^b	N/R	117.9 ^a
city11_b	282.1 ^a	441.8 ^b	N/R	300.1 ^b	N/R	N/R

Sources: ^a[6], ^b[5], ^c[4].

Table 4: Absolute trajectory error (ATE, cm) on VECtor for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [3]	ESIO [3]	DEIO [4]	ES-PTAM [1]
corner_slow	1.06 ^a	2.15 ^b	1.43 ^a	2.67 ^b	10.70 ^a	N/R
robot_normal	3.16 ^a	4.81 ^b	4.95 ^a	5.17 ^b	N/R	N/R
desk_normal	6.48 ^a	16.47 ^b	6.00 ^a	N/R	254.8 ^a	N/R
sofa_normal	7.18 ^a	40.28 ^b	5.06 ^a	43.94 ^b	N/R	N/R
hdr_normal	6.71 ^a	13.53 ^b	1.96 ^a	27.85 ^b	N/R	N/R
corridors_dolly	196.5 ^a	N/R	92.04 ^a	N/R	492.6 ^d	N/R
units_dolly	444.5 ^a	N/R	872.1 ^a	N/R	826.4 ^d	N/R

Sources: ^a[6], ^b[5], ^d[2].

Table 5: Absolute trajectory error (ATE, cm) on TUM-VIE for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [3]	ESIO [3]	DEIO [4]	ES-PTAM [1]
tumvie_1d_trans	1.05 ^a	3.33 ^b	N/R	N/R	1.85 ^a	1.05 ^b
tumvie_3d_trans	1.26 ^a	7.26 ^b	N/R	N/R	1.28 ^a	8.53 ^b
tumvie_6dof	1.69 ^a	3.21 ^b	N/R	N/R	1.45 ^a	10.25 ^b
tumvie_desk	2.44 ^a	6.16 ^b	N/R	N/R	1.49 ^a	2.50 ^b
tumvie_desk2	1.79 ^a	4.02 ^b	N/R	N/R	1.45 ^a	7.20 ^b

Sources: ^a[6], ^b[5].

Table 6: Absolute trajectory error (ATE, cm) on Other for stereo/metric event odometry baselines, taken from the literature. All values are SE(3)-aligned full-trajectory results; monocular up-to-scale (Sim(3)) results are excluded as they are not the same quantity. Where several papers report the same method and sequence we quote the best published value. Superscripts give the source of each number, which is not always the paper proposing the method. N/R marks a cell no surveyed source reports; such cells must be produced by us or carry a reason code.

Sequence	Stereo-DEVO [6]	ESVO2 [5]	ESVIO [3]	ESIO [3]	DEIO [4]	ES-PTAM [1]
hnu_campus	151.8 ^a	181.8 ^a	N/R	1420.5 ^a	N/R	N/R
drone_fast	13.29 ^a	N/R	N/R	154.0 ^a	N/R	N/R

Sources: ^a[6].